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In [32]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
file_path = "EDA sheet.csv"
df = pd.read_csv(file_path)

# Get basic information about the dataset
print(df.info())

# Check for missing values
print(df.isnull().sum())

# Q1: Maximum matches played by a player in a season
# Find the player with the most matches
max_matches = df["Matches"].max()
player_max_matches = df[df["Matches"] == max_matches][["Player", "Matches"]]

# Print the result
print(player_max_matches)

# Q2: Top 2 players with highest average who have at least 2 half-centuries
top_avg_players = df[df["50"] >= 2].nlargest(2, "Avg")[["Player", "Avg"]]
print(top_avg_players)

# Q3: Create first and last name columns
df[["First Name", "Last Name"]] = df["Player"].str.split(n=1, expand=True)

# Q4: Clean 'Highest Score' column
df['Cleaned_Highest_Score'] = df['Highest Score'].astype(str).str.replace

# Check the cleaned column
print(df[['Highest Score', 'Cleaned_Highest_Score']].head())

# Q5: Total number of centuries scored
total_centuries = df[df["100"] > 0]["100"].sum()
print(f"Total Centuries: {total_centuries}")

# Q6: Players with strike rate less than average strike rate
average_strike_rate = df["Strike rate"].mean()
low_sr_players = df[df["Strike rate"] < average_strike_rate][["Player", "Strike rate"]]
print(low_sr_players)

# Q7: Correlation heatmap
plt.figure(figsize=(10, 6))

# Select only numeric columns for correlation matrix
numeric_df = df.select_dtypes(include=['number'])

sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
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plt.title("Feature Correlation Heatmap")
plt.show()

# Q8: Players with average > 50 and strike rate > 120
high_performance_players = df[(df["Avg"] > 50) & (df["Strike rate"] > 120)]
print(high_performance_players[["Player", "Avg", "Strike rate"]])

# Q9: Players with avg > 40 and balls faced > 100
consistent_players = df[(df["Avg"] > 40) & (df["Balls faced"] > 100)]
print(consistent_players[["Player", "Avg", "Balls faced"]])

# Q10: Players with at least one century
century_players = df[df["100"] >= 1][["Player"]]
plt.figure(figsize=(8, 4))
sns.countplot(y=century_players, order=century_players.value_counts().index)
plt.title("Players with at least one century")
plt.show()

# Q11: Players with at least 4 half-centuries
half_century_players = df[df["50"] >= 4][["Player"]]
print(half_century_players)

# Q12: Players with more than 45 boundaries and 10 sixes
boundary_six_players = df[(df["4s"] > 45) & (df["6s"] > 10)][["Player", "Runs"]]
print(boundary_six_players)

# Q13: Histogram of matches played
plt.hist(df["Matches"], bins=10, color='skyblue', edgecolor='black')
plt.xlabel("Matches Played")
plt.ylabel("Frequency")
plt.title("Distribution of Matches Played by Players")
plt.show()

# Q14: Histogram of balls faced
plt.hist(df["Balls faced"], bins=10, color='orange', edgecolor='black')
plt.xlabel("Balls faced")
plt.ylabel("Frequency")
plt.title("Distribution of Balls faced by Players")
plt.show()

# Q15: Top 10 players with most runs
top_run_scorers = df.nlargest(10, "Runs")[["Player", "Runs"]]
print(top_run_scorers)

# Q16: Players who played but didn't bat
non_batting_players = df[df["Inns"] == 0][["Player"]]
print(non_batting_players)

# Q17: Percentage of total runs from 4s and 6s
df["Runs_from_Boundaries"] = df["4s"]*4 + df["6s"]*6
df["Boundary_Percentage"] = (df["Runs_from_Boundaries"] / df["Runs"]) * 100
top_boundary_players = df.nlargest(5, "Boundary_Percentage")[["Player", "Runs"]]
print(top_boundary_players)

# Q18: Top 5 players with highest Not Out %
df["Not_Out_Percentage"] = (df["Not Out"] / df["Inns"]) * 100
top_not_out_players = df.nlargest(5, "Not_Out_Percentage")[["Player", "Not Out"]]
print(top_not_out_players)

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# Q19: Top 10 players with most sixes
plt.figure(figsize=(8, 4))
sns.barplot(x=df.nlargest(10, "6s")["6s"], y=df.nlargest(10, "6s")["Player"])
plt.xlabel("Number of Sixes")
plt.ylabel("Player")
plt.title("Top 10 Players with Most Sixes")
plt.show()

# Q20: Scatter plot of runs vs balls faced
plt.figure(figsize=(8, 6))
sns.scatterplot(x=df["Balls faced"], y=df["Runs"], alpha=0.7)
plt.xlabel("Balls faced")
plt.ylabel("Runs Scored")
plt.title("Runs vs Balls faced")
plt.show()

# Checking correlation between Runs and Balls faced
correlation = df["Balls faced"].corr(df["Runs"])
print(f"Correlation between Balls faced and Runs: {correlation:.2f}")
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 133 entries, 0 to 132
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Jersey No             133 non-null   int64
1   Player                133 non-null   object
2   Matches               133 non-null   int64
3   Inns                 133 non-null   int64
4   Not Out               133 non-null   int64
5   Runs                  133 non-null   int64
6   Highest Score         133 non-null   object
7   Avg                   133 non-null   float64
8   Balls faced           133 non-null   int64
9   Strike rate           133 non-null   float64
10  100                   133 non-null   int64
11  50                    133 non-null   int64
12  4s                    133 non-null   int64
13  6s                    133 non-null   int64
```

dtypes: float64(2), int64(10), object(2)

memory usage: 14.7+ KB

```
Jersey No      0
Player         0
Matches        0
Inns           0
Not Out        0
Runs           0
Highest Score  0
Avg            0
Balls faced    0
Strike rate    0
100            0
50             0
4s             0
6s             0
```

dtype: int64

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      Player  Matches
1  Shikhar Dhawan    17
3    Shreyas Iyer    17
19 Marcus Stoinis    17
67 Kagiso Rabada     17
```

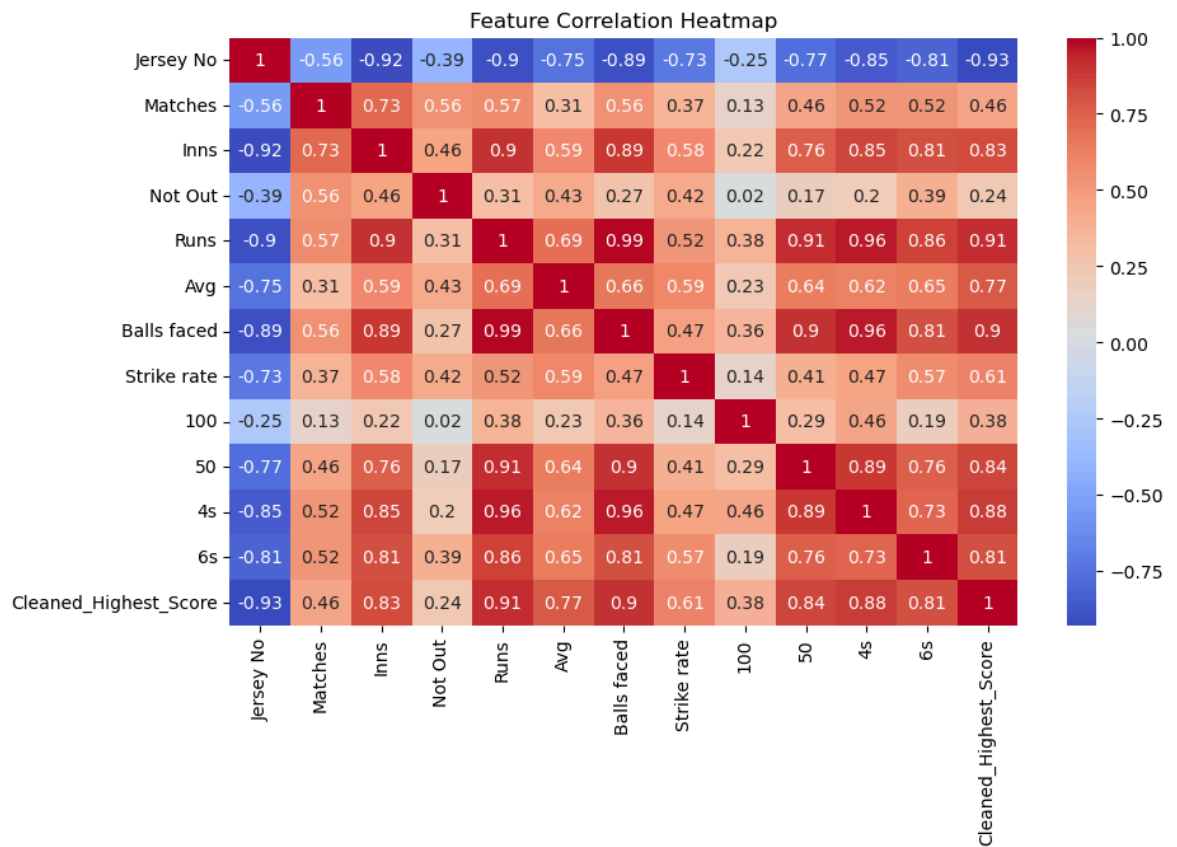
```
      Player  Avg
36 Wriddhiman Saha  71.33
4   Ishan Kishan  57.33
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  Highest Score  Cleaned_Highest_Score
0           132*                132
1           106*                106
2            85*                 85
3            88*                 88
4             99                 99
```

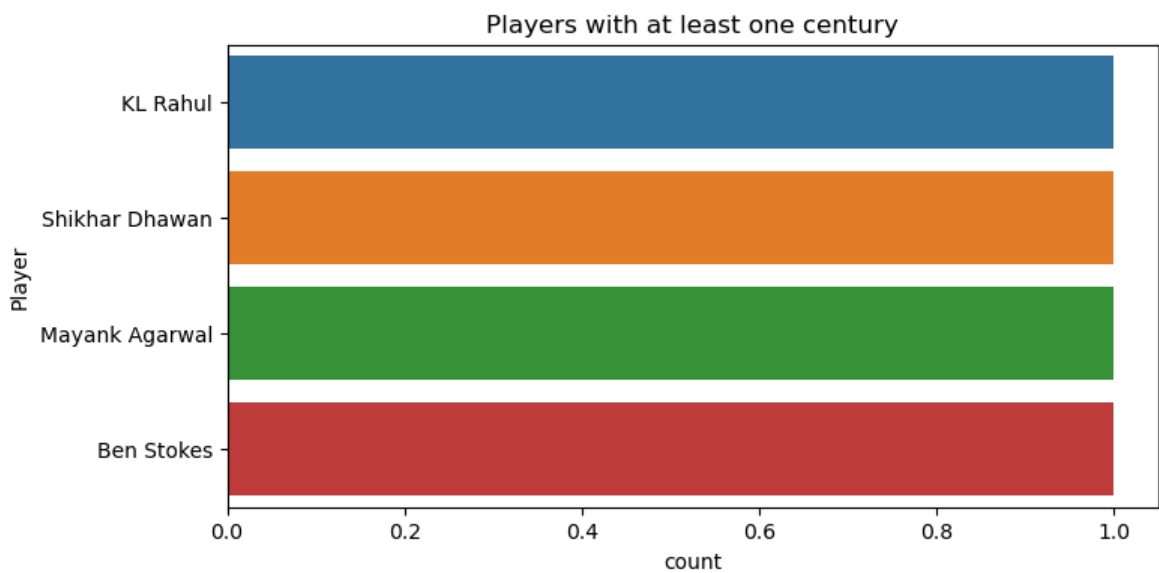
Total Centuries: 5

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      Player  Strike rate
51  Ajinkya Rahane    105.60
55   Glenn Maxwell    101.88
58   Vijay Shankar    101.04
61   Josh Philippe    101.29
62  Gurkeerat Singh     88.75
65   Kedar Jadhav     93.93
70  Yashasvi Jaiswal    90.90
71   Shreyas Gopal     94.87
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77	Murali Vijay	74.41
79	Chris Jordan	93.54
80	Navdeep Saini	100.00
82	Kamlesh Nagarkoti	70.96
84	Harshal Patel	87.50
85	Jimmy Neesham	105.55
86	Tom Banton	90.00
89	Prabhsimran Singh	100.00
92	Kuldeep Yadav	61.90
94	Moeen Ali	75.00
95	Sandeep Sharma	80.00
96	Shardul Thakur	57.14
98	Rinku Singh	100.00
99	Shivam Mavi	71.42
100	Varun Chakaravarthy	66.66
101	Jaydev Unadkat	69.23
102	Ankit Rajpoot	90.00
104	Shahbaz Nadeem	87.50
105	Pravin Dubey	53.84
107	Deepak Chahar	58.33
108	Ravi Bishnoi	58.33
109	Andrew Tye	100.00
111	Kartik Tyagi	66.66
112	Murugan Ashwin	100.00
114	T Natarajan	60.00
115	Prasidh Krishna	50.00
116	Rahul Chahar	50.00
117	Mohammad Shami	66.66
118	Nikhil Naik	33.33
119	Mujeeb Ur Rahman	33.33
120	Dale Steyn	50.00
121	Varun Aaron	10.00
122	Shahbaz Ahmed	100.00
123	Yuzvendra Chahal	33.33
124	Mitchell Marsh	0.00
125	Umesh Yadav	0.00
126	Bhuvneshwar Kumar	0.00
127	Sheldon Cottrell	0.00
128	Khaleel Ahmed	0.00
129	Arshdeep Singh	0.00
130	Daniel Sams	0.00
131	Shreevats Goswami	0.00
132	Trent Boult	0.00



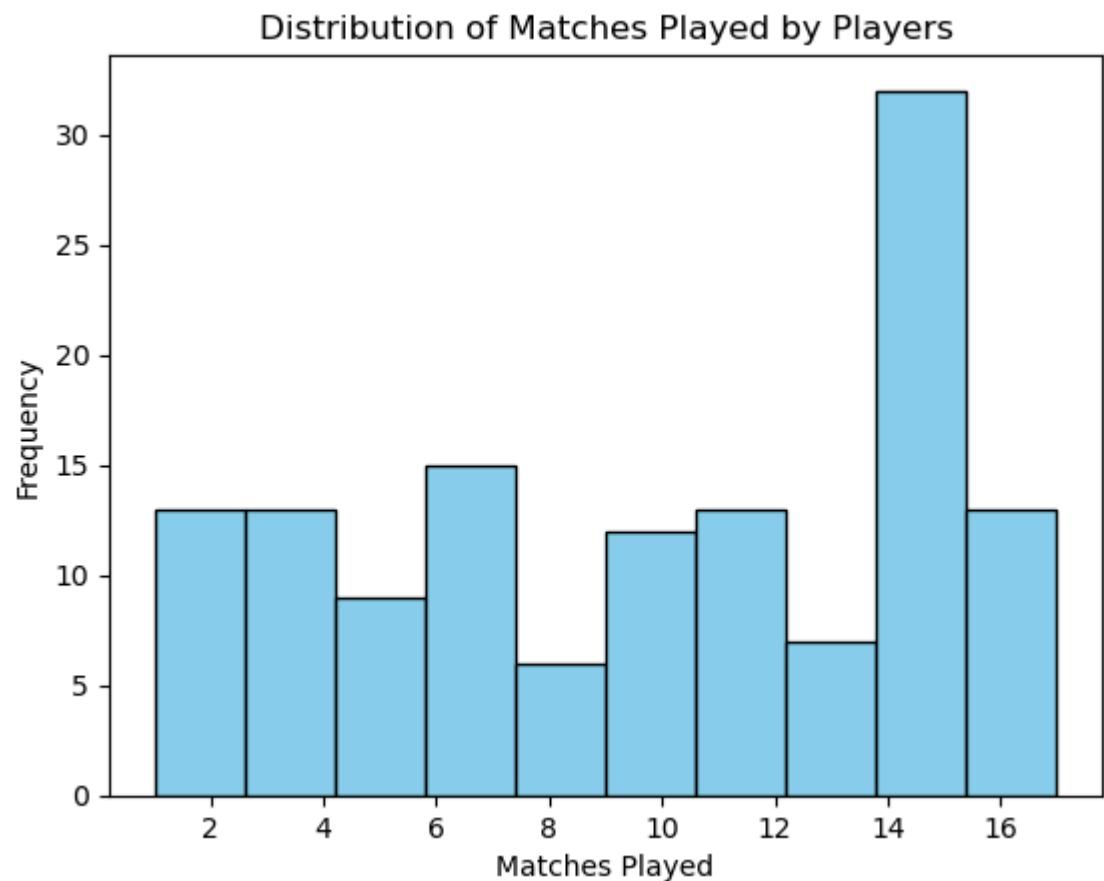
	Player	Avg	Strike rate
0	KL Rahul	55.83	129.34
4	Ishan Kishan	57.33	145.76
31	Kieron Pollard	53.60	191.42
36	Wriddhiman Saha	71.33	139.86
37	Ruturaj Gaikwad	51.00	120.71
57	Deepak Hooda	101.00	142.25
60	Tom Curran	83.00	133.87
	Player	Avg	Balls faced
0	KL Rahul	55.83	518
1	Shikhar Dhawan	44.14	427
4	Ishan Kishan	57.33	354
8	Virat Kohli	42.36	384
9	ABD Villiers	45.40	286
10	Faf Duplessis	40.81	319
14	Eoin Morgan	41.80	302
24	Kane Williamson	45.28	237
27	Chris Gayle	41.14	210
28	Ben Stokes	40.71	200
31	Kieron Pollard	53.60	140
32	Rahul Tewatia	42.50	183
33	Ravindra Jadeja	46.40	135
36	Wriddhiman Saha	71.33	153
37	Ruturaj Gaikwad	51.00	169



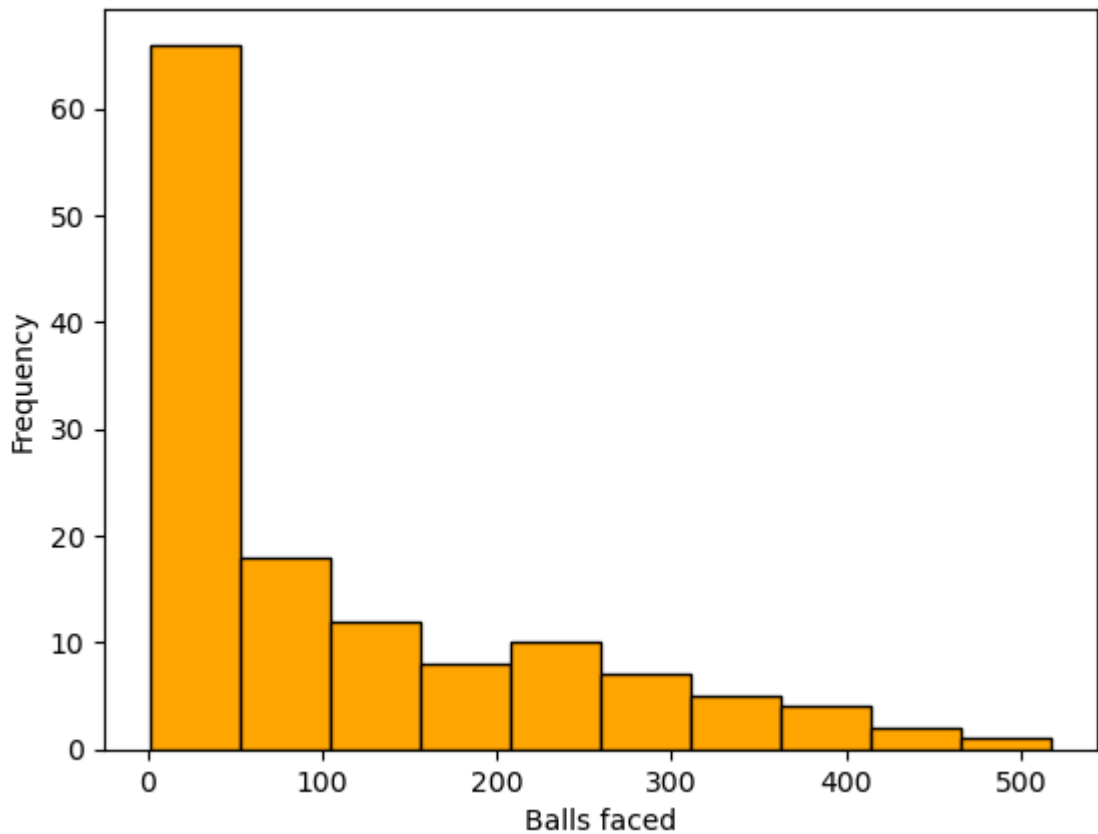
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0      KL Rahul
1      Shikhar Dhawan
2      David Warner
4      Ishan Kishan
5      Quinton Kock
6      Suryakumar Yadav
7      Devdutt Padikkal
9      ABD Villiers
10     Faf Duplessis
Name: Player, dtype: object
      Player  4s  6s
0      KL Rahul  58  23
1  Shikhar Dhawan  67  12
2      David Warner  52  14
5      Quinton Kock  46  22
6  Suryakumar Yadav  61  11

```



Distribution of Balls faced by Players

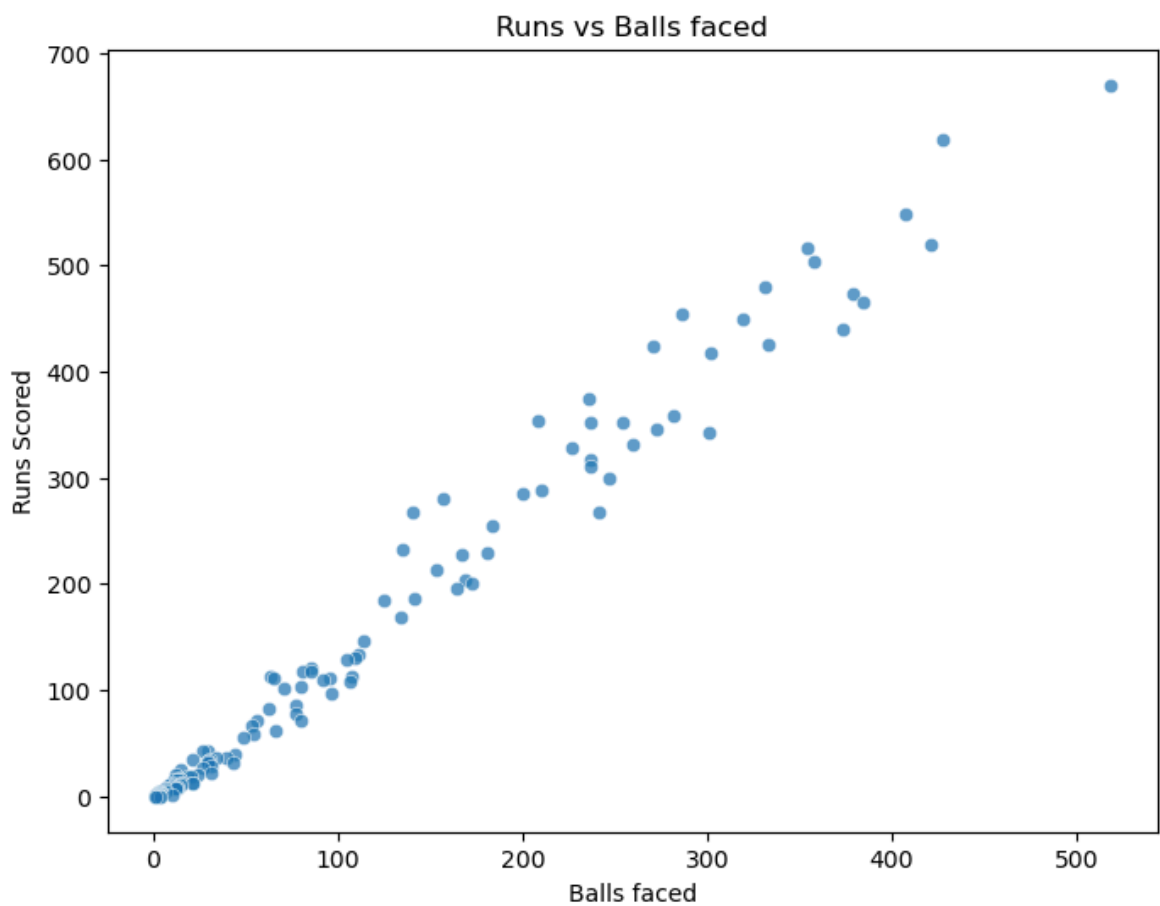
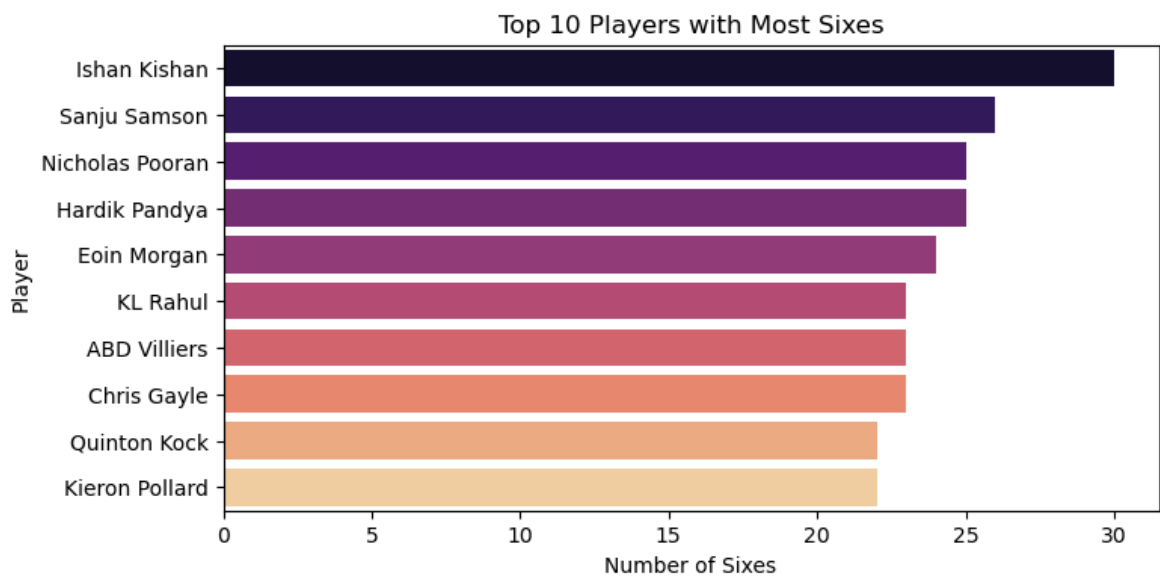


	Player	Runs
0	KL Rahul	670
1	Shikhar Dhawan	618
2	David Warner	548
3	Shreyas Iyer	519
4	Ishan Kishan	516
5	Quinton Kock	503
6	Suryakumar Yadav	480
7	Devdutt Padikkal	473
8	Virat Kohli	466
9	ABD Villiers	454

Series([], Name: Player, dtype: object)

	Player	Boundary_Percentage
109	Andrew Tye	100.000000
48	Andre Russell	76.923077
74	Chris Morris	76.470588
29	Hardik Pandya	73.309609
47	Sunil Narine	72.727273

	Player	Not_Out_Percentage
62	Gurkeerat Singh	100.0
68	Lockie Ferguson	100.0
93	Imran Tahir	100.0
97	Mohammad Nabi	100.0
105	Pravin Dubey	100.0



Correlation between Balls faced and Runs: 0.99

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